Research Statement

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I study how the gains from trade respond to government policies, climate change, and infrastructure. Most of my projects share a common approach. I start from a frontier quantitative trade model, adapt it just enough to speak to the policy question at hand, and take it to data to measure what a policy or a shock does to real income across countries or regions. The models are structural, so I can study counterfactuals that have never been observed, such as a world without trade agreements or a river closed to barge traffic. My job market paper applies this approach to the world trading system as a whole, and several of my other projects apply it to agricultural trade, where trade is regulated more than in most sectors, data are rich, and the answers matter for food security.

The value of the world trading system

My job market paper asks what the world trading system is worth. Tariffs today sit far below the levels governments would choose on their own, and the system holds them there in two ways. Governments are bound by negotiated tariff ceilings, and they weigh the welfare of their trading partners. I build a model in which both forces operate and use observed tariffs to measure how much each one matters. Removing both would lower world real income by close to one percent. Either one alone preserves most of that value, and large and small economies rely on different parts of the system. I then ask how the system's value changes as comparative advantage shifts, including under projected climate change. The answer depends on which countries and sectors a shock touches, not on its sign alone, and the tariff ceilings carry most of the response. The framework is a general tool for valuing trade rules under alternative futures, and I plan to use it next to study what the system is worth when a large member stops honoring its ceilings.

Infrastructure and the geography of trade

The Mississippi River carries a large share of US grain exports, and keeping it navigable costs the federal government money every year. In a sole-authored paper, I ask what the river is worth as a trade artery. I build a spatial general equilibrium model of the US economy with transport costs computed on a fine grid and close the river to barge traffic in the model. Real income falls by about a third of a percent, close to thirty billion dollars a year, which is several times the cost of keeping the river open. I am revising the paper after a first round of referee reports.

Regulation, technology, and agricultural trade

With Sandro Steinbach, Carlos Zurita, and Felipe Barbosa, I study how regulation shapes the spread of agricultural technology across countries and what that spread is worth. Genetically modified crops raise yields where they are grown, but many countries ban their cultivation or their

import, and the bans change trade patterns and the incentive to adopt elsewhere. We extend a quantitative model of agricultural trade to include this regulation and measure what liberalization, a global ban, and new traits in wheat and rice would do to real income and land use. A companion paper uses the same model with projected climate data to ask how much of the income loss from climate change biotechnology can offset. A third project applies machine learning to predict when countries approve new traits. The USDA supports this work through a grant on the economics of biotechnology regulation.

Agricultural trade policy in practice

Two further projects bring the tools closer to the instruments policymakers actually use. With Yu Lwin, Sandro Steinbach, Carlos Zurita, and Shawn Arita, I study who along the US beef supply chain bears the tariffs of 2025 and 2026, combining a trade model with pass-through elasticities estimated at each stage from the border to the retail counter. The paper is under revise and resubmit at *Agricultural Economics*. In a sole-authored paper in progress, I show how to bring tariff-rate quotas, which govern trade in sugar, dairy, beef, and rice across much of the world, into standard quantitative trade models that so far treat every tariff as a single wedge. The theory is complete and the empirical application is under way.

Looking ahead

My agenda joins these strands in two directions. The first is the value of trade rules under a changing climate and a changing geography of production. The second is the measurement of agricultural policy with the right instruments in the model. I aim to publish in general-interest and field journals such as the *Journal of International Economics*, the *American Economic Journal: Applied Economics*, and the *American Journal of Agricultural Economics*, and I will continue to write for policy audiences, as in a recent article in *Choices*.